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Cpts 360
PA4

I implemented my proxy by first listening for connecting clients on a provided port given at the command line. Then, once a connection is established, I fork the program to allow for multiple connections. In the child process, I call my `handle_request()` function, and close any unnecessary file descriptors. The parent process simply closes the connection file descriptor, as it is not needed, and then continues listening for other connections.

In the `handle_request()` function, I initialize my RIO buffer and then retrieve the method, uri, and version. If the method isn't a GET method, I return. Otherwise, I call my `forward_request()` function which will send the request to the appropriate server.

In my `forward_request()` function, I send the uri information to my `parse_uri` function, which extracts the hostname, the port number, and the query information. I then open a connection with the port and hostname information using `open_clientfd()`. I then construct an appropriate GET request using my `make_header()` function. The `make_header()` function takes the hostname and query information and creates a header to send to the server. It concatenates all the necessary headers to send, such as "Connection: close\r\n". It also checks for any additional headers that need to be sent, if they were sent from the server upon connection.

After the headers are created and set, I write the headers to the server using `Rio_writen()` with the server file descriptor. Then, I tell the server I am sending no more transmissions. I use a loop and `Rio_readlineb()` to read the transmitted data from the server, and send the data back to the client using `Rio_writen()` with the client file descriptor. Finally, I close the server file descriptor. When this function returns, the client file descriptor is closed in main.

Why can't we access some URLs using our proxy?

Most URLs use HTTPS, which is an encrypted connection. Our simple proxy does not handle the CONNECT method or tunneling, which is necessary for HTTPS requests. Essentially, our simple proxy can only handle GET requests, where we're retrieving raw, unencrypted data from a server and forwarding it back to the client. Since we didn't set anything up to handle the CONNECT method, it won't work.